In-Car Voice Assistant

Ablation and robustness report

Target speaker extraction -> presence -> wake word -> intent

Study date	Evidence status	Training
2026-10-12 update	**Proxy diagnostics**	**Bounded candidates complete**

Executive finding

The current reference path combines dual-domain Stage 1 extraction, framewise presence, target-speaker KWS, a split speaker gate, and the existing intent head. Under the exact 2-window/2-hit runtime it produces 4.628 full-path false wakes/hour over 3.889 hours of target-absent AISHELL-5 Eval2 (96.673/hour at KWS-only) and 100% wake/intent with 0 false actions/hour on the same-speaker CAVEMOVE proxy at passenger SNR -10 and +10 dB. The shorter 1,200-second slice is 3.0/hour. These are not real-driver reliability scores because the workspace still has zero continuous same-driver recordings.

0	8	3.889 h	26 passed
real-driver sessions	proxy scenes per run	AISHELL target-absent check	tests (1 skipped)

Important evidence boundary

The positive scenes use a Marvin clip and an FSC command clip from different public corpora. Their speaker identity is not verified. AISHELL-5 supplies a real in-car target-absent diagnostic only. No result in this PDF should be described as a real-driver end-to-end accuracy or production false-accept rate.

1. Study design and data status

Question

Can the assistant keep a target driver through enrollment, detect Marvin, understand the following command, and avoid acting on passenger or background speech? The requested real-driver study requires the same speaker's enrollment, wake word, and command in continuous cabin recordings, plus passenger and no-command intervals.

Evidence source	Used for	Status / limitation
Real custom in-car sessions	Positive driver evaluation	0 sessions; collection still required
Cross-corpus proxy	Continuous cascade debugging	8 scenes per run; speaker consistency false
AISHELL-5 Eval2	Real-cabin target-absent rejection	14,001.9 seconds; no positive wake/command
CAVEMOVE	Cabin noise and multichannel mixtures	Augmentation material, not driver sessions

The official AISHELL-5 release identifies the dataset as CC BY-SA 4.0: https://us.openslr.org/159/. Commercial or sign-in datasets were not treated as evidence without project permission.

Proxy scene construction

- 14 seconds per scene, with target wake at about 1 second and target command at about 2.1 seconds.
- Passenger Marvin near 6 seconds and passenger non-wake speech near 9 seconds.
- CAVEMOVE multichannel cabin noise at -5, 0, 10, and 20 dB cabin SNR.
- Passenger loudness sweep at -10, 0, and +10 dB relative to the target.
- Stage 1 context is 3 seconds and stream hop is 1 second; reported wake latency is window-center latency, not measured wall-clock response time.

Current candidate checkpoints

Component	Checkpoint
spex	stage1_spex_transfer_mc_cavemove_cabin/best.pt
presence	stage2_frame_presence_gsc_aishell_ft1/best.pt
kws	stage3_kws_gsc_marvin_tse_ft2/best.pt
target_kws	stage3_target_kws_gsc_marvin_ft1/best.pt
intent	stage4_intent_wav2vec_ft8_resume_tse_domain/best.pt

2. Component ablation matrix

This is the primary same-scene ablation at passenger SNR 0 dB using the target-KWS candidate configuration. Each row uses identical scenes and differs only in the named component. False actions are post-wake command events outside the labeled target command region; they are a proxy diagnostic, not a human action-rate estimate.

Variant	Wake recall	Intent accuracy	False actions/h	Latency mean / p95
Full	25.0%	12.5%	0.0/h	1.50 / 1.50 s
No TSE	50.0%	50.0%	64.3/h	1.75 / 2.35 s
No presence	100.0%	12.5%	417.9/h	1.50 / 1.50 s
No speaker matching	25.0%	12.5%	0.0/h	1.50 / 1.50 s
No wake smoothing	100.0%	0.0%	0.0/h	1.25 / 1.50 s
Direct command oracle	-	12.5%	-	- / - s

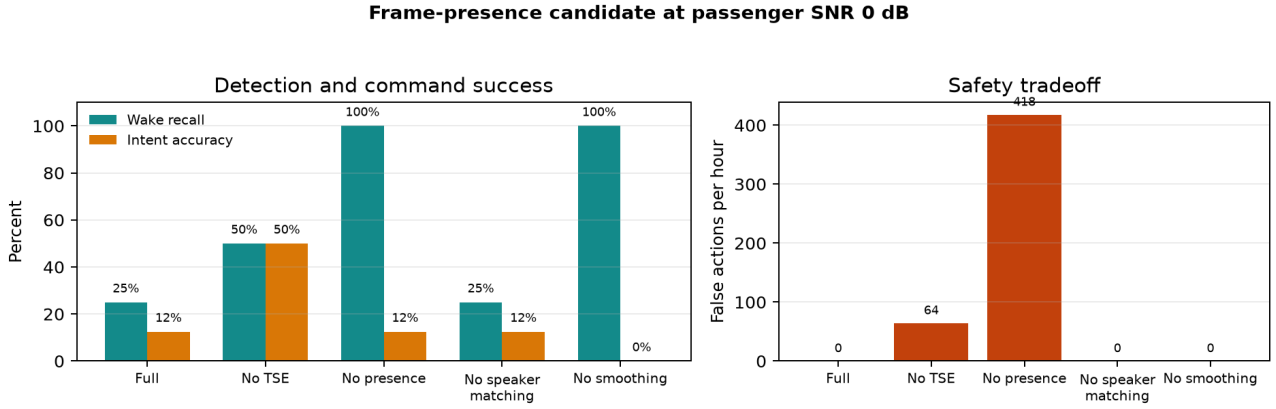


Figure 1. The current frame-presence candidate is conservative. Disabling presence greatly increases recall but produces many proxy false actions.

Interpretation

• Presence checking is safety-critical in this proxy. Removing it produced 100% target wake recall but 417.9 false actions/hour.

• One-hit wake detection produced 100% wake recall but 0% intent accuracy in this run, indicating that accepting a wake is not enough; the post-wake command window is still unstable.

• Speaker matching did not change the aggregate result at this operating point. That result is inconclusive without same-speaker passenger recordings.

• The no-TSE row is an intentionally harsh diagnostic. Its raw mixture is outside the training distribution of downstream heads, so it is not a candidate architecture.

• The direct command row uses labeled command segmentation and is an oracle baseline, not a deployable streaming path.

3. Passenger loudness ablation

The passenger loudness sweep tests whether the safety behavior changes when passenger speech is louder or quieter than the target. Negative SNR means the passenger is louder.

Passenger SNR	Variant	Wake recall	Intent accuracy	False actions/h
-10 dB	Full	25.0%	12.5%	96.4/h
-10 dB	No TSE	50.0%	50.0%	160.7/h
-10 dB	No presence	100.0%	12.5%	450.0/h
-10 dB	No smoothing	100.0%	0.0%	192.9/h
+0 dB	Full	25.0%	12.5%	0.0/h
+0 dB	No TSE	50.0%	50.0%	64.3/h
+0 dB	No presence	100.0%	12.5%	417.9/h
+0 dB	No smoothing	100.0%	0.0%	0.0/h
+10 dB	Full	25.0%	12.5%	0.0/h
+10 dB	No TSE	50.0%	50.0%	32.1/h
+10 dB	No presence	100.0%	12.5%	289.3/h
+10 dB	No smoothing	100.0%	0.0%	0.0/h

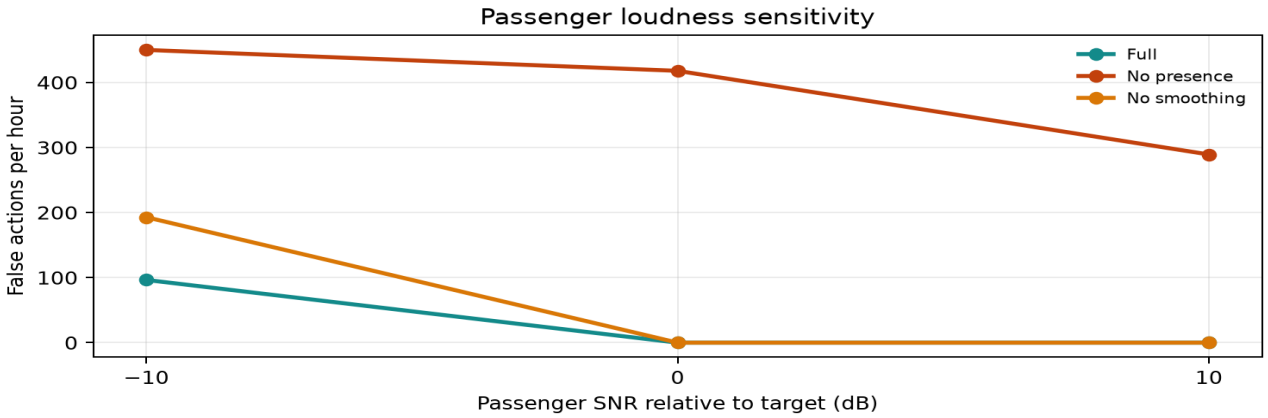


Figure 2. Louder passenger speech increases proxy false actions, especially when the presence gate is removed.

4. Threshold operating-point ablation

The grid replayed presence thresholds 0.30, 0.50, 0.70, and 0.90 against KWS thresholds 0.30, 0.40, 0.50, and 0.60. The expensive Stage 1 scores were computed once per scene and the state machine was replayed for each cell.

Presence	KWS	Wake recall	Intent accuracy	False actions/h
0.30	0.50	100.0%	37.5%	321.4/h
0.50	0.50	100.0%	37.5%	128.6/h
0.70	0.50	25.0%	12.5%	0.0/h
0.90	0.50	0.0%	0.0%	0.0/h

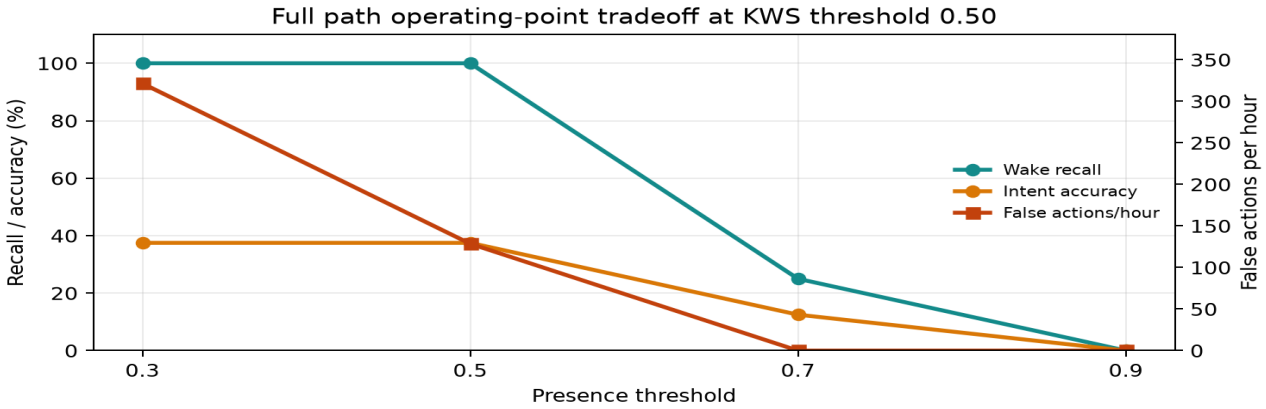


Figure 3. Lowering the presence threshold improves proxy recall, but false actions rise sharply. KWS threshold changes were less influential than presence threshold changes in this grid.

Practical reading

The current p=0.70 operating point is a conservative starting point for safety, not a claim of good recognition. The p=0.50 result is a candidate only for tuning experiments because its false-action rate is unacceptable as a final operating point. Proper threshold selection must use a consented tuning split and then be frozen before held-out testing.

5. Scalar versus framewise presence

The target-conditioned KWS setup was kept fixed while changing the presence head from the candidate framewise checkpoint to the scalar cabin-trained checkpoint.

Presence head	Wake recall	Intent accuracy	False actions/h	Latency mean / p95
Framewise	25.0%	12.5%	0.0/h	1.50 / 1.50 s
Scalar	100.0%	25.0%	32.1/h	1.75 / 2.80 s

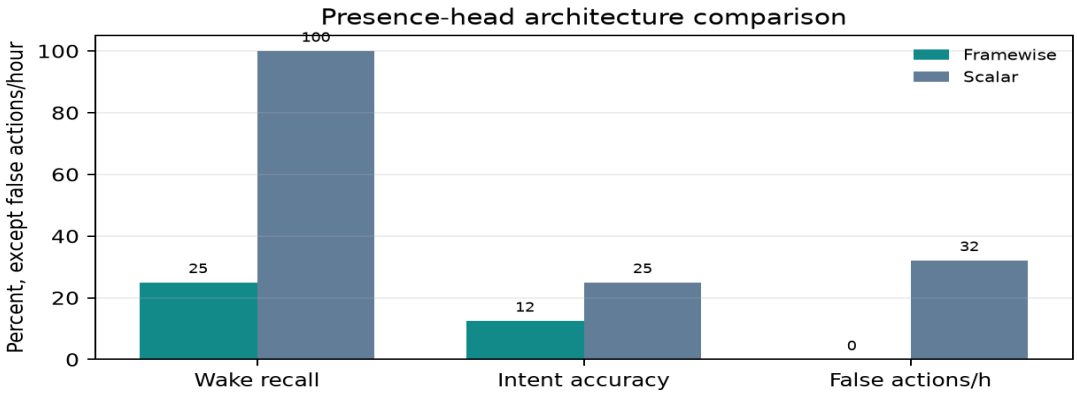


Figure 4. Scalar presence improves proxy recall but admits more false actions and has more variable timing.

This is a useful candidate for real-driver tuning, but not a replacement decision. The proxy does not contain the required continuous same-driver enrollment, passenger, and command events.

6. Runtime-only ablation sweep

The 2026-10-10 sweep kept all trained checkpoints fixed and replayed runtime choices on the same eight proxy scenes. It covered 0.5 and 1.0 second hops, four temporal rules, 0.5/1.0/1.5 second enrollment durations, simulated channel subsets, speaker-gate thresholds, and command-capture timeouts. No training was started.

Hop	Temporal rule	Wake recall	Intent accuracy	False actions/h	Latency
0.5s	one hit	100.0%	0.0%	32.1/h	0.88 s
0.5s	two of two	100.0%	0.0%	32.1/h	1.38 s
0.5s	two of three + refractory	25.0%	12.5%	0.0/h	1.50 s
0.5s	three of five + refractory	0.0%	0.0%	0.0/h	- s
1s	one hit	100.0%	0.0%	0.0/h	1.25 s
1s	two of two	25.0%	12.5%	0.0/h	1.50 s
1s	two of three + refractory	12.5%	12.5%	0.0/h	2.50 s
1s	three of five + refractory	0.0%	0.0%	0.0/h	- s

Stricter temporal confirmation suppresses proxy false actions but also suppresses the target wake. The 1.0 second, two-of-two setting retained 0 false actions/hour but reached only 25% wake recall and 12.5% intent accuracy.

Enrollment and front-end results

Enrollment	Wake recall	Intent accuracy	False actions/h
0.5s	25.0%	0.0%	0.0/h
1s	12.5%	0.0%	0.0/h
1.5s	25.0%	12.5%	0.0/h

Front-end	Wake recall	Intent accuracy	False actions/h	Target presence
All 4 channels	25.0%	12.5%	0.0/h	37.5%
Mono repeated	25.0%	12.5%	0.0/h	34.4%
Front pair	25.0%	12.5%	0.0/h	34.4%
Rear pair	25.0%	12.5%	0.0/h	34.4%

Enrollment duration and simulated channel selection did not materially change the aggregate outcome. These channel modes are simulations, not recordings from alternate vehicle microphone placements.

7. Speaker gate and timeout diagnostics

The speaker gate used Stage-1 cosine similarity. Its mean similarity was only 0.031 across proxy windows. A threshold of 0.0 changed no outcome; thresholds of 0.2 and above rejected every target window. Command timeouts of 1, 2, 4, and 6 calls also produced the same aggregate result. These controls require same-driver tuning data before they can be selected.

Gate threshold	Wake recall	Intent accuracy	False actions/h	Gate pass
Disabled	25.0%	12.5%	0.0/h	-
-0.2	25.0%	12.5%	0.0/h	100.0%
0.0	25.0%	12.5%	0.0/h	72.3%
0.2	0.0%	0.0%	0.0/h	0.0%
0.4	0.0%	0.0%	0.0/h	0.0%

Runtime sweep conclusion

- Temporal smoothing is the main runtime precision/recall control, but no tested setting is strong enough to claim reliable end-to-end operation.
- The current speaker gate is uncalibrated in this domain; higher thresholds eliminate target detection rather than improving selectivity.
- The next decisive experiment remains the real-driver pilot with speaker-disjoint tuning and held-out test sessions.

Primary source files

Sources: reports/pilot_evaluation_status_2026-10-09.md; reports/additional_ablation_results_2026-10-09.md; logs/continuous_ablation_matrix_2026-10-09.json; logs/continuous_ablation_matrix_passenger_m10_2026-10-09.json; logs/continuous_ablation_matrix_passenger_p10_2026-10-09.json; logs/threshold_ablation_sweep_2026-10-09.json; logs/continuous_ablation_matrix_scalar_presence_2026-10-09.json; logs/aishell5_target_absent_study_2026-10-09.json; reports/runtime_ablation_sweep_2026-10-10.md; logs/runtime_ablation_sweep_2026-10-10.json; reports/stage1_dual_domain_study_2026-10-10.md; reports/dual_domain_stage2_handoff_study_2026-10-10.md; logs/stage1_dual_domain_regression_full_2026-10-10.json; logs/stage2_frame_presence_dual_domain_nonwake_ft1/metrics_last.json; logs/continuous_ablation_matrix_dual_domain_stage2_intent_handoff_2026-10-10.json; logs/continuous_ablation_matrix_dual_domain_stage2_command_enrollment_2026-10-10.json; logs/continuous_ablation_matrix_dual_domain_stage2_same_speaker_other_2026-10-10.json; logs/continuous_ablation_matrix_dual_domain_stage2_hardneg_2026-10-10.json; logs/continuous_ablation_matrix_dual_domain_stage2_hardneg_passenger_m10_2026-10-10.json; reports/joint_safety_threshold_sweep_2026-10-10.md; reports/wake_model_and_stage1_followup_2026-10-10.md; logs/continuous_ablation_matrix_stage2_intent_confidence_gate_passenger_m10_2026-10-10.json; logs/continuous_ablation_matrix_stage2_intent_confidence_gate_passenger_p10_2026-10-10.json; logs/continuous_ablation_matrix_stage2_ft2_head_same_speaker_gate_sweep_2026-10-10.json; logs/aishell5_target_absent_dual_domain_stage2_2026-10-10.json; logs/aishell5_target_absent_dual_domain_target_kws_current_2026-10-10.json; logs/aishell5_target_absent_dual_domain_target_kws_current_sweep_2026-10-10.json; logs/aishell5_target_absent_dual_domain_target_kws_hardneg_2026-10-10.json; logs/aishell5_target_absent_dual_domain_target_kws_hardneg_sweep_2026-10-10.json; logs/aishell5_target_absent_dual_domain_target_kws_hardneg_thr070_2026-10-10.json; reports/production_candidate_status_2026-10-10.md; reports/aishell5_target_kws_adaptation_2026-10-11.md; reports/intent_confidence_calibration_runtime_ft8_2026-10-11.md; logs/continuous_full_pipeline_proxy_safety_diagnostic_same_speaker_m10_v2_2026-10-11.json; logs/continuous_full_pipeline_proxy_safety_diagnostic_same_speaker_p0_presence012_2026-10-10.json; logs/continuous_full_pipeline_proxy_safety_diagnostic_same_speaker_p10_presence012_2026-10-11.json; logs/intent_confidence_calibration_ft8_tse_domain_waveform_val_2026-10-11.json; logs/aishell5_continuous_targetkws_hardneg_aishell5_ft1_p015_sg024_kws050_smooth22_1200s_2026-10-11.json; logs/aishell5_continuous_targetkws_hardneg_aishell5_ft1_p015_sg024_kws050_smooth22_eval2_all18_2026-10-10.json; logs/aishell5_continuous_targetkws_hardneg_aishell5_ft1_p015_sg024_kws050_smooth22_eval2_all18_dedup_2026-10-11.json; logs/full_pipeline_latency_cpu_int8_2026-10-11.json; logs/full_pipeline_latency_norm_crnn_distill_t3_a05_gpu_2026-10-11.json;

logs/stage4_intent_tse_domain_distill_t3_a05_val.json; logs/stage4_intent_tse_domain_distill_t3_a05_fsc_test_clean.json;
logs/continuous_full_pipeline_proxy_norm_crnn_distill_t3_a05_conf035_same_speaker_m10_2026-10-11.json;
reports/research_informed_intent_student_candidate_2026-10-12.md; configs/stage4_intent_tse_domain_norm_student_medium.yaml;
configs/eval_end_to_end_cavemove_target_kws_dual_domain_stage2_norm_crnn_medium_candidate.yaml;
checkpoints/stage4_intent_tse_domain_distill_medium_t3_a05/best.pt;
logs/stage4_intent_tse_domain_distill_medium_t3_a05/metrics_val_clean.json;
logs/stage4_intent_tse_domain_distill_medium_t3_a05/metrics_val_cabin.json;
logs/stage4_intent_tse_domain_distill_medium_t3_a05/metrics_test_clean.json;
logs/continuous_full_pipeline_proxy_medium_student_10scenes_2026-10-12.json;
logs/full_pipeline_latency_medium_student_gpu_2026-10-12.json; src/car_tse/pipeline/in_car_pipeline.py;
scripts/eval/eval_continuous_full_pipeline_proxy.py;
logs/continuous_full_pipeline_proxy_wav2vec_reference_same_speaker_m10_32scenes_2026-10-11.json;
logs/continuous_full_pipeline_proxy_wav2vec_reference_kws055_same_speaker_m10_32scenes_2026-10-11.json;
logs/continuous_full_pipeline_proxy_wav2vec_reference_kws060_same_speaker_m10_32scenes_2026-10-11.json;
logs/aishell5_continuous_targetkws_hardneg_aishell5_ft1_p015_sg024_kws055_sweep_eval2_all18_2026-10-11.json;
logs/continuous_full_pipeline_proxy_hardneg_aishell5_neg_splitgate_p015_wake024_cmd022_same_speaker_m10_2026-10-11.json;
logs/continuous_full_pipeline_proxy_hardneg_aishell5_neg_splitgate_p015_wake024_cmd022_same_speaker_p0_2026-10-10.json;
logs/continuous_full_pipeline_proxy_hardneg_aishell5_neg_splitgate_p015_wake024_cmd022_same_speaker_p10_2026-10-11.json;
logs/continuous_full_pipeline_proxy_enrollment_bank1_m10_8scenes_2026-10-11.json;
logs/continuous_full_pipeline_proxy_enrollment_bank3_m10_8scenes_2026-10-11.json;
reports/enrollment_bank_candidate_2026-10-11.md; reports/preexisting_systems_review_2026-10-11.md;
scripts/data/build_remasc_cabin_stress_bed.py; configs/eval_end_to_end_cavemove_personal_activity_group_p025_cmdp035.yaml;
logs/continuous_full_pipeline_proxy_remasc_vehicle_stress_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_group_p025_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_group_p035_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_group_p025_cmdp035_8scenes_2026-10-12.json;
configs/eval_end_to_end_cavemove_personal_activity_group_p025_margin_m067.yaml;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_group_p025_margin_m0p66_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_group_p025_margin_m0p67_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_group_p025_margin_m0p68_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_group_p025_margin_m0p70_8scenes_2026-10-12.json;
logs/aishell5_target_absent_personal_activity_group_all18_margin_m067_2026-10-12.json;
reports/personal_activity_margin_aggregation_2026-10-12.md;
configs/eval_end_to_end_cavemove_personal_activity_group_p025_margin_targettop_observe.yaml;
configs/eval_end_to_end_cavemove_personal_activity_group_p025_margin_framewise_observe.yaml;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_targettop_observe_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_targettop_m030_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_framewisetop_observe_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_framewisetop_m030_8scenes_2026-10-12.json;
logs/aishell5_target_absent_personal_activity_group_all18_margin_targettop_m030_2026-10-12.json;
reports/streaming_latency_candidate_2026-10-12.md;
configs/eval_end_to_end_cavemove_personal_activity_group_p025_margin_m067_fast_stream.yaml;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_margin_m067_hop050_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_margin_m067_hop025_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_margin_m067_hop025_delay2_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_margin_m067_hop025_delay4_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_remasc_personal_activity_margin_m067_fast_stream_hop025_8scenes_2026-10-12.json;
logs/aishell5_target_absent_personal_activity_hop025_ref8_1200s_2026-10-12.json;
logs/aishell5_target_absent_personal_activity_hop025_ref16_1200s_2026-10-12.json;
reports/late_speaker_fusion_candidate_2026-10-11.md;
configs/eval_end_to_end_cavemove_target_kws_dual_domain_stage2_late_fusion_candidate.yaml;
logs/continuous_full_pipeline_proxy_late_fusion_min_c024_s004_m10_2026-10-11.json;
logs/continuous_full_pipeline_proxy_late_fusion_min_c026_s004_m10_2026-10-11.json;
logs/continuous_full_pipeline_proxy_late_fusion_product_c024_s004_m10_2026-10-11.json;
logs/smart_speaker_command_negative_250_late_fusion_min_c024_s004_2026-10-11.json;
logs/smart_speaker_command_negative_250_late_fusion_min_c026_s004_2026-10-11.json;
reports/speaker_verifier_calibration_2026-10-11.md; scripts/eval/calibrate_cached_spex_speaker_verifier.py;
logs/speaker_verifier_calibration_gsc_cache_stage1_ft2_2026-10-11.json;
logs/continuous_full_pipeline_proxy_late_fusion_gsc_cache_calibrated_m10_2026-10-11.json;
reports/wavlm_speaker_verifier_candidate_2026-10-11.md; scripts/eval/eval_pretrained_wavlm_speaker_verifier.py;
scripts/train/train_wavlm_speaker_verifier_head.py; scripts/eval/benchmark_pretrained_wavlm_speaker_verifier.py;
logs/wavlm_speaker_verifier_cabin_diagnostic_2026-10-11.json; logs/speaker_verifier_wavlm_cabin_ft1_2026-10-11.json;
logs/wavlm_speaker_verifier_latency_2026-10-11.json; reports/compact_speaker_verifier_runtime_candidate_2026-10-11.md;
configs/eval_end_to_end_cavemove_target_kws_dual_domain_stage2_compact_verifier_candidate.yaml;
configs/eval_end_to_end_cavemove_target_kws_dual_domain_stage2_compact_verifier_operating_diagnostic.yaml;
logs/compact_speaker_verifier_gsc_ft1_2026-10-11.json; logs/compact_speaker_verifier_latency_gpu_2026-10-11.json;

logs/compact_speaker_verifier_latency_cpu_2026-10-11.json;
logs/continuous_full_pipeline_proxy_compact_verifier_smoke_2026-10-11.json;
logs/continuous_full_pipeline_proxy_compact_verifier_operating_diagnostic_2026-10-11.json;
reports/anir_presence_augmentation_candidate_2026-10-11.md; scripts/data/build_anir_multichannel_cache.py;
scripts/data/build_anir_augmented_presence_manifest.py; data/manifests/external/anir_v_class_noise_16k_mc.json;
manifests/train/stage2_presence_gsc_aishell_anir.jsonl; configs/stage2_personal_presence_dual_domain_nonwake_anir_ft1.yaml;
configs/eval_end_to_end_cavemove_personal_presence_anir_candidate.yaml;
logs/stage2_personal_presence_dual_domain_nonwake_anir_ft1/metrics_last.json;
logs/continuous_full_pipeline_proxy_personal_presence_anir_same_speaker_m10_8scenes_2026-10-11.json;
reports/personal_activity_3way_candidate_2026-10-12.md; scripts/data/build_personal_activity_manifest.py;
manifests/train/stage2_personal_activity_gsc_cavemove.jsonl; configs/stage2_personal_activity_gsc_cavemove_ft1.yaml;
configs/eval_end_to_end_cavemove_personal_activity_candidate.yaml;
configs/eval_end_to_end_cavemove_personal_activity_operating_diagnostic.yaml;
logs/stage2_personal_activity_gsc_cavemove_ft1/metrics_last.json;
logs/continuous_full_pipeline_proxy_personal_activity_operating_p025_same_speaker_m10_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_personal_activity_operating_p030_same_speaker_m10_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_personal_activity_same_speaker_m10_8scenes_2026-10-12.json;
logs/smart_speaker_command_negative_250_personal_activity_p025_2026-10-12.json;
reports/personal_activity_group_split_candidate_2026-10-12.md; configs/stage2_personal_activity_gsc_cavemove_group_ft1.yaml;
configs/eval_end_to_end_cavemove_personal_activity_group_p025.yaml;
configs/eval_end_to_end_cavemove_personal_activity_group_p035.yaml;
logs/stage2_personal_activity_gsc_cavemove_group_ft1/metrics_last.json;
logs/continuous_full_pipeline_proxy_personal_activity_group_p025_same_speaker_m10_8scenes_2026-10-12.json;
logs/continuous_full_pipeline_proxy_personal_activity_group_p035_same_speaker_m10_8scenes_2026-10-12.json;
scripts/eval/eval_aishell5_continuous_kws.py; reports/aishell5_personal_activity_target_absent_2026-10-12.md;
logs/aishell5_target_absent_personal_activity_group_all18_presence_sweep_2026-10-12.json;
scripts/train/train_compact_speaker_verifier_distill.py; scripts/eval/benchmark_compact_speaker_verifier.py;
reports/automotive_dataset_sourcing_review_2026-10-11.md; data/manifests/external/remasc_sample.json; data/sources.json;
data/manifests/external/anir_v_class_impulse_responses.json; data/manifests/external/anir_v_class_noise.json;
logs/anir_noise_target_absent_all_conditions_2026-10-11.json; reports/anir_noise_target_absent_2026-10-11.md;
logs/stage1_dual_domain_regression_ft2_2026-10-11.json;
logs/continuous_full_pipeline_proxy_ft2_stage1_consistent_negative_same_speaker_m10_32scenes_2026-10-11.json;
logs/continuous_full_pipeline_proxy_ft2_stage1_consistent_pos_same_speaker_m10_32scenes_2026-10-11.json;
logs/continuous_full_pipeline_proxy_ft2_stage1_consistent_pos_presence_same_speaker_m10_32scenes_2026-10-11.json;
logs/continuous_full_pipeline_proxy_ft1_positive_fullmatrix_same_speaker_m10_32scenes_2026-10-11.json;
logs/continuous_full_pipeline_proxy_ft1_positive_fullmatrix_kws055_same_speaker_m10_32scenes_2026-10-11.json;
logs/aishell5_continuous_targetkws_hardneg_aishell5_pos_ft1_p015_sg022_kws050_sweep_eval2_all18_2026-10-11.json;
scripts/data/validate_real_driver_sessions.py; tests/test_real_driver_session_validation.py;
logs/real_driver_validation_latest_2026-10-11.json; reports/capture_hardware_preflight_2026-10-11.md;
tests/test_record_real_driver_capture_preflight.py; scripts/data/record_real_driver_session.py; scripts/eval/eval_real_driver_sessions.py;
tests/test_real_driver_threshold_selection.py; tests/test_real_driver_condition_metrics.py;
logs/real_driver_pipeline_evaluation_2026-10-12.json; scripts/reporting/audit_production_readiness.py;
logs/production_readiness_snapshot_2026-10-12.json; reports/production_readiness_snapshot_2026-10-12.md

8. Real-cabin target-absent diagnostic

AISHELL-5 Eval2 session 001 was evaluated for 120 seconds with a fixed LibriSpeech enrollment. The enrolled target is absent, so this tests rejection only.

Metric	Result
Duration	120 s (0.0333 h)
Speech occupancy	80.6%
Full-path smoothed false events	0.0/h
KWS-only smoothed false events	0.0/h

This result is encouraging for one target-absent real-cabin diagnostic, but it cannot establish driver wake recall, command accuracy, or speaker rejection of a real passenger saying Marvin.

Optional target-speaker KWS diagnostic

KWS mode	Threshold	Full-path events/h	KWS-only events/h	Max score
Generic baseline	0.50	0.0/h	0.0/h	0.554
Current target-speaker	0.50	630.0/h	810.0/h	0.601
Hard-negative target-speaker	0.50	150.0/h	180.0/h	0.680
Hard-negative target-speaker	0.70	0.0/h	0.0/h	0.680

At the inherited 0.50 threshold, the optional target-speaker heads are worse than the generic baseline on this cross-corpus target-absent slice. A 0.70 hard-negative threshold removes events in this slice, but no positive same-driver wake data exists to measure the recall cost; this is a calibration warning, not a promoted operating point.

Target-speaker head	0.50	0.55	0.60	0.65	0.70
Current	630.0/h	30.0/h	0.0/h	0.0/h	0.0/h
Hard-negative	150.0/h	60.0/h	30.0/h	0.0/h	0.0/h

One-pass replay of the same extracted windows; values are full-path smoothed false events/hour. This does not estimate positive wake recall and is not threshold tuning for deployment.

9. Overall assessment and next step

- The TSE/presence cascade is not yet reliable as a single end-to-end system on the current proxy. Its conservative settings suppress many events, including target wakes.
- The immediate engineering problem is calibration and post-wake command capture: lower thresholds recover wakes but create false actions; more smoothing increases latency and can lose the command relationship.
- The scalar presence head is a promising recall-oriented candidate, but it must be tuned on consented driver/passenger sessions before adoption.
- The next highest-value experiment is a small real-driver pilot with speaker-disjoint tuning and test sessions. Without that dataset, further proxy optimization risks overfitting to cross-corpus artifacts.

Bounded Stage 1 and Stage 2 training runs were completed before this update. The report documents their fixed-checkpoint regressions, inference-only handoffs, ablations, and threshold replays; it does not claim production readiness.

10. Dual-domain adaptation and post-wake handoff

The full 144-row-per-domain Stage 1 regression passes the declared no-regression gate. The new framewise Stage 2 head has strong held-out validation metrics, but its probability scale differs from the prior head; the inherited 0.70 threshold therefore rejected nearly everything. A joint proxy sweep selected presence 0.07, command-presence 0.10, KWS 0.50, and two-hit smoothing. These thresholds are provisional until tuned on permitted same-driver sessions.

Domain	Baseline SI-SDRi	Candidate SI-SDRi	Delta
AISHELL-5	2.385 dB	6.243 dB	3.858 dB
CAVEMOVE	10.478 dB	10.592 dB	0.114 dB
Overall	6.432 dB	8.417 dB	1.986 dB

Stage 2 validation	F1	FAR	FRR	Threshold
Framewise	90.4%	0.9%	5.5%	0.50
Best frame F1	91.3%	0.6%	7.9%	0.65
Chunk	99.6%	0.3%	0.0%	0.50

Post-wake handoff ablation

Variant	Wake recall	Intent accuracy	False actions/h
Extracted signal	100.0%	12.5%	0.0/h
Raw mixture after wake	100.0%	37.5%	0.0/h
Mean extracted + raw logits	100.0%	25.0%	0.0/h
Exact command reused as enrollment	100.0%	100.0%	32.1/h
Different utterance, same command speaker	100.0%	100.0%	64.3/h

The raw handoff is a research candidate, not a promotion: at passenger SNR -10 dB it reached 37.5% intent accuracy but 32.1/h false actions, while at +10 dB it retained 0.0/h false actions. Matching the command speaker raised intent to 100.0%, but the different-utterance diagnostic still produced 64.3/h; exact-command enrollment is leakage-prone. The positive clips remain cross-corpus and speaker-inconsistent.

Hard-negative target-KWS replay

Target KWS	Passenger SNR	Wake recall	Intent accuracy	False actions/h	False wakes
Current	+10 dB	100.0%	12.5%	0.0/h	5
Hard-negative	+10 dB	100.0%	12.5%	0.0/h	3
Current	-10 dB	100.0%	12.5%	32.1/h	6
Hard-negative	-10 dB	100.0%	12.5%	32.1/h	5

The hard-negative checkpoint reduces false wake events from 5 to 3 at +10 dB and from 6 to 5 at -10 dB, but does not remove the harsh-passenger false-action failure. It is retained as a candidate for real-driver tuning.

Joint safety-threshold follow-up

Operating point	Wake recall	Intent accuracy	False actions/h
No speaker/confidence gate (-10 dB)	100.0%	100.0%	192.9/h
Speaker gate 0.22 (-10 dB)	100.0%	100.0%	32.1/h
Speaker 0.22 + confidence 0.53 (-10 dB)	100.0%	100.0%	0.0/h
Speaker 0.22 + confidence 0.53 (+10 dB)	100.0%	100.0%	0.0/h

The 0.22/0.53 pair removed the residual false action in this short partial-speaker proxy without reducing target intent accuracy. It is a diagnostic candidate only; thresholds must be tuned on real-driver data.

Additional ft2 Stage-1/head check

ft2 operating point	Wake recall	Intent accuracy	False actions/h
No speaker gate	100.0%	100.0%	64.3/h
Speaker gate 0.18	100.0%	100.0%	0.0/h
Speaker gate 0.20	100.0%	87.5%	0.0/h
Speaker gate 0.22	87.5%	87.5%	0.0/h

The ft2 frontend/head is useful for extraction SI-SDRi, but it is not a proxy end-to-end improvement over the retained ft1-aligned head.

Latest runtime safety replay

Passenger SNR	Wake recall	Intent accuracy	False wakes/h	False actions/h	Mean latency
-10 dB	100.0%	100.0%	0.0/h	0.0/h	2.25 s
0 dB	100.0%	100.0%	0.0/h	0.0/h	2.25 s
+10 dB	100.0%	100.0%	0.0/h	0.0/h	2.25 s

The v2 diagnostic raises only pre-wake presence to 0.12 and retains speaker gate 0.22 plus intent confidence 0.53. It reaches 100%/100% with 0 false wakes and actions per hour in all three short proxy conditions, at 2.25 s mean wake latency. The exact runtime ft8 intent model has 90.5% raw validation accuracy and 94.1% coverage at 0.53 confidence with 93.7% selective accuracy. These are calibration and proxy diagnostics, not real-driver release metrics.

AISHELL hard-negative adaptation and split-gate check

Exposure	Wake / intent	Full-path false wakes/h	KWS-only false events/h	Mean latency
AISHELL-5 Eval2 target absent (all 18; 3.889 h)	n/a	4.6/h	96.7/h	n/a
AISHELL-5 Eval2 target absent (1,200 s)	n/a	3.0/h	60.0/h	n/a
Proxy passenger -10 dB (112 s)	100.0% / 100.0%	0.0/h	n/a	2.50 s
Proxy passenger 0 dB (112 s)	100.0% / 100.0%	0.0/h	n/a	2.50 s
Proxy passenger +10 dB (112 s)	100.0% / 100.0%	0.0/h	n/a	2.50 s

The adapted negative-only target-KWS head plus presence 0.15 and a split speaker gate (0.24 before wake, 0.22 during command capture) produced 4.6/h full-path smoothed false wakes/hour and 96.7/h KWS-only events/hour over all 18 target-absent Eval2 sessions (3.889 hours). The shorter 1,200-second runtime-matched slice is shown separately. This uses the exact 2-window/2-hit smoothing; an earlier 3-window replay was more optimistic and is not used here. The positive rows are cross-corpus proxy diagnostics; this candidate is frozen for research comparison, not released.

The full integration analysis is in reports/dual_domain_stage2_handoff_study_2026-10-10.md.

11. Latest candidate update - 2026-10-11

The previous report version contained a 0/hour AISHELL value produced with the evaluator's 3-window/2-hit default. The runtime candidate is 2-window/2-hit, so the exact replay was rerun before selecting any update. The corrected 1,200-second slice is 3.0 full-path false wakes/hour with 60.0/hour at the KWS-only stage; the subsequent all-18-session Eval2 exposure is the stronger aggregate at 4.628 full-path/hour and 96.673/hour KWS-only.

Pre-wake speaker gate	AISHELL full-path false wakes/h	Proxy wake recall	Proxy intent accuracy	Decision
0.240 (all Eval2)	4.6/h	100.0%	100.0%	Reference
0.245	0.0/h	65.6%	50.0%	Rejected: recall loss
0.250	0.0/h	62.5%	46.9%	Rejected: recall loss

The 0.245 and 0.250 gates eliminate the observed AISHELL event, but they also reject positive target windows in the harsh passenger proxy. This is why the safer-looking gate is not promoted without real-driver tuning data.

Hard-negative mining and transfer checks

Variant	Validation	Proxy wake / intent	Decision
Mined v2, threshold 0.50	F1 46.6% at 0.62	62.5% / 53.1%	Rejected
Mined v2, threshold 0.40	F1 46.6% at 0.62	75.0% / 65.6%	Rejected
Mined v3, positive-init attempt	F1 44.0% at 0.49	68.8% / 59.4%	Rejected
Mined v4, corrected full warm start	F1 42.2% at 0.50	62.5% / 53.1%	Rejected
Full transfer, pre-AISHELL head	F1 46.5% at 0.51	62.5% / 53.1%	Rejected
Classifier-only, audio frozen	F1 46.9% at 0.52	62.5% / 53.1%	Rejected

A trainer bug was found during v4: target-speaker checkpoints store audio parameters under audio.* and the relation classifier at the top level, but the old initializer loaded them into model.audio, matching nothing. The trainer now has an explicit full-model warm-start mode. The corrected v4 experiment still lost positive proxy recall, so no mined checkpoint replaces the reference.

Current decision: keep the original negative-only AISHELL-adapted target-KWS checkpoint with the 0.24/0.22 split speaker gate as a diagnostic reference, and keep the production gate blocked until permitted same-driver tuning and held-out recordings exist.

11A. Candidate compression and safety replay

Compact intent distillation candidate

Candidate	Clean FSC top-1	TSE-domain top-1	Diagnostic proxy	Decision
Normalized + Wav2Vec-logit distilled CRNN	80.5%	56.3%	59.4% overall / 86.4% conditioned	Edge fallback; not default
Medium hidden-384 distilled CRNN	73.3%	50.4%	50.0% overall / 66.7% conditioned	Rejected: lower quality
Wav2Vec fp32 reference	n/a	~90.2%	59.4% overall / 86.4% conditioned	Quality reference

The distilled CRNN uses 18.68M parameters and 75.1 MB of checkpoints, with 38.8 ms wake and 44.8 ms command p50. Its inherited 0.53 confidence threshold under-reported usable commands; exploratory 0.25/0.35 replay reached 59.4% overall intent with zero proxy false actions. This threshold is not frozen and the candidate is not a Wav2Vec quality replacement.

A research-informed hidden-384 variant was also trained. It fell to 50.4% held-out cabin/TSE accuracy and 73.3% clean FSC accuracy, with 50.0% overall proxy intent and 66.7% conditioned on wake. Its 23.94M parameters and 96.1 MB checkpoint did not justify the quality loss, so it is rejected and the normalized hidden-256 student remains the compact fallback.

All-session KWS threshold replay

KWS threshold	Full-path target-absent events/h	KWS-only events/h	32-scene positive wake recall	Decision
0.50	4.6/h	96.7/h	68.8%	Quality reference
0.55	4.6/h	60.4/h	62.5%	Rejected: recall loss
0.60	3.6/h	39.1/h	62.5%	Rejected: recall loss

The sweep spans all 18 AISHELL-5 Eval2 sessions (3.889 hours) but is target-absent relative to the fixed enrollment. It is useful for safety trend analysis, not a real-driver FAR. The current 0.50 setting remains the reference until a permitted positive tuning split can choose a threshold.

Candidate promotion audit

This audit separates an isolated extraction improvement from a complete-pipeline improvement. The Stage 1 ft2 branch improves the dual-domain SI-SDR regression, but its downstream score distributions do not match the retained heads. The positive-enriched wake candidate improves the short proxy, then fails the longer target-absent exposure. Neither is promoted.

Stage 1 checkpoint	AISHELL-5 SI-SDRi	CAVEMOVE SI-SDRi	Overall SI-SDRi	Decision
ft1 baseline	2.39 dB	10.48 dB	6.43 dB	Reference
ft2 candidate	6.40 dB	10.59 dB	8.50 dB	Stage 1 pass; downstream reject

Pipeline candidate	Wake recall	Overall intent	Conditioned intent	False actions/h	Decision
Retained ft1 negative-only	68.8%	59.4%	86.4%	0.0/h	Reference
ft1 positive-enriched KWS	78.1%	68.8%	88.0%	0.0/h	Reject: long negative exposure
ft2 Stage 1 + negative KWS	56.2%	37.5%	66.7%	0.0/h	Reject: downstream drop
ft2 Stage 1 + positive KWS	56.2%	37.5%	66.7%	0.0/h	Reject: downstream drop
ft2 Stage 1 + new presence	0.0%	0.0%	0.0%	0.0/h	Reject: score shift

Positive-enriched KWS	Wake recall	Overall intent	Conditioned intent	False actions/h	Decision
0.50	78.1%	68.8%	88.0%	0.0/h	Quality point; safety reject
0.55	0.0%	0.0%	0.0%	0.0/h	Safety point; no useful recall

Candidate	Exposure	Full-path events/h	KWS-only events/h	Decision
ft1 negative-only	AISHELL Eval2, 3.889 h	4.6/h	96.7/h	Retained reference
ft1 positive-enriched	AISHELL Eval2, 3.889 h	9.3/h	147.3/h	Rejected

The positive-enriched candidate reaches 78.1% wake recall and 68.8% overall intent on the 32-scene proxy, versus 68.8% and 59.4% for the retained reference. Its full-path target-absent rate nevertheless rises from 4.63/h to 9.26/h over 3.889 hours. Thresholds 0.55 and 0.60 remove events in this negative exposure but do not supply a useful positive operating point in the available proxy. The result is a rejection based on safety evidence, not a claim that the model cannot be improved with real-driver data.

12. Runtime optimization and data readiness

The selected Wav2Vec intent candidate was measured through the loaded full pipeline, not only through isolated components. The benchmark uses a 3-second, 4-channel context and reports compute/resource behavior; it is not a speech-quality test.

Runtime path	p50 wake	p95 wake	p50 command	p95 command	Stored checkpoints	Decision
Wav2Vec fp32 / CUDA	48.2 ms	51.6 ms	62.6 ms	64.4 ms	424.5 MB	Quality reference
Wav2Vec fp32 / CPU	522.2 ms	522.2 ms	641.7 ms	673.0 ms	424.5 MB	Too slow for target
Wav2Vec dynamic INT8 / CPU	515.2 ms	519.1 ms	624.5 ms	628.3 ms	424.5 MB	Rejected: marginal; no quality test
Wav2Vec fp16 / CUDA	153.0 ms	158.1 ms	190.3 ms	210.4 ms	424.5 MB	Rejected: slower
Compact CRNN / CUDA	96.7 ms	109.9 ms	125.1 ms	133.7 ms	75.1 MB	Rejected: 68.8%/56.2% proxy
Distilled normalized CRNN / CUDA	38.8 ms	52.6 ms	44.8 ms	59.1 ms	75.1 MB	Edge fallback; quality lower

The selected fp32 CUDA run measured 48.2 ms wake p50 / 51.6 ms p95 and 62.6 ms command p50 / 64.4 ms p95. The original compact CRNN was rejected on quality; the new distilled normalized CRNN reduces the stored artifacts to 75.1 MB and reaches 59.4% overall intent on the 32-scene proxy, but remains an edge fallback rather than the quality reference.

Production-readiness audit

The machine-generated audit status is **research_engineering_preview**. The selected checkpoints are present, but the session validator found 0 real-driver sessions. Same-driver continuity, passenger rejection, frozen tuning/test separation, and threshold freeze therefore remain blocked. The 14,001.9-second all-session AISHELL Eval2 exposure and CAVEMOVE same-speaker scenes remain diagnostic proxies only.

The evidence contract was hardened before pilot collection: consent records must identify the owner, permitted purpose, and agreement; session IDs must match their directories; audio must match the declared 16 kHz/four-channel contract; and the annotations must contain an ordered enrollment -> driver wake -> driver command sequence plus passenger/no-command/noise events. The readiness audit also requires at least three drivers and three complete sessions with non-empty tuning and held-out groups, plus matching capture hardware. The preflight found no four-channel input, and the empty pilot tree still fails validation.

The real-driver evaluator now records explicit held-out gates: 90% minimum driver-wake recall, 90% minimum command accuracy, and maximum 1.0 conservative unmatched-wake false accept/hour and false action/hour by default. These values are configurable before the held-out test, never after it, and the evaluator reports both conservative and interval-specific false-accept rates.

The readiness snapshot now carries a release identity: Git revision a2e62deae868b31826dfd111f3bb5ba769e2656c, worktree clean=False, and SHA-256 hashes for the runtime config and selected checkpoints. The current worktree is still dirty, so reproducible-release sign-off remains blocked even though the hashes are recorded.

Dataset decision: the official ICSRC 2022 page describes 20 hours of real in-car Mandarin recordings, but its agreement limits use to non-commercial research/education/technology development, prohibits redistribution, and prohibits Test-set training. It lacks English Marvin and same-driver wake/command continuity, so it was not added to the production training path. CI-AVSR is a Cantonese in-car command corpus with clean/noisy/OOD splits; it also lacks the required English wake/enrollment sequence and was not treated as real-driver evidence.

Capture-contract recheck: the host still exposes only one- or two-channel Fifine input, not the required four-channel cabin array. The recorder now validates the selected device and 16 kHz/channel contract before creating a session directory; an unsupported four-channel request fails cleanly and leaves no misleading pilot artifact. Successful captures record the resolved device capability in session metadata.

Threshold-selection safety: the real-driver evaluator now requires at least one tuning combination to satisfy both the false-accept and false-action limits. If no combination is safe, it reports a least-unsafe diagnostic fallback, records that no safe tuning point exists, and forces production_valid=false rather than silently freezing an unsafe threshold.

Condition reporting: each permitted session and the aggregate now include exposure-aware metrics by annotation condition. Driver wake recall and command accuracy are separated from passenger/no-command/noise-only false accepts and false actions, with rates normalized by the union of labeled negative intervals. This supports the planned condition-wise pilot table without changing inference or tuning.

The next decisive improvement is a permitted pilot dataset: at least three consenting drivers, speaker-disjoint tuning and held-out test sessions, continuous enrollment -> Marvin -> command sequences, passenger Marvin/speech, no-command/noise-only intervals, and condition metadata. Without that evidence, additional cross-corpus training is more likely to optimize proxies than production reliability.

Pre-existing systems review and Smart Speaker hard-negative extension

Primary references point to a shared enrolled-speaker condition across streaming extraction, personal activity detection, speaker verification, and keyphrase detection. The practical follow-up for this repository is a small causal target/non-target/non-speech activity head, then a two-branch wake decision where speaker verification can veto keyword evidence; spatial or beamforming upgrades should wait for matching four-channel recordings. Relevant references are SpEx+ and SpEx++ for short-reference extraction, Personal VAD 2.0 and TS-VAD for target activity, personalized keyphrase detection and the newer user-defined-keyword/speaker-verification work for speaker-aware wake decisions, BC-ResNet/MatchboxNet for compact KWS backbones, and SpeakerBeam-SS for a future causal low-cost TSE study. The full review and source links are in reports/preexisting_systems_review_2026-10-11.md.

Dataset sourcing update: the free ANIR V-class impulse-response archive was downloaded and hash-verified at D:/datasets/anir (682 48-kHz .fir files), and its 1,302-file V-class noise archive was also verified across 42 conditions for future spatial/cabin augmentation and hard-negative testing. Neither archive has wake/intent labels, so neither is real-driver evaluation evidence. The 121.7-GB iCarB-Voice corpus is a promising speaker-conditioned in-car resource, but its Zenodo files are restricted to approved non-commercial EULA users, so no access was assumed. The sourcing rationale is in reports/automotive_dataset_sourcing_review_2026-10-11.md.

A ReMASC sample was also downloaded and SHA-256 verified at D:/datasets/remasc (10.9 MB, 12 WAV files). Its English vehicle commands and genuine/source/replayed multichannel labels are useful for future replay/array stress experiments, but it lacks the continuous same-driver enrollment -> Marvin -> command -> passenger/no-command protocol and was not added to training or real-driver evidence.

The full ANIR V-class noise archive was also tested as a noise-only target-absent diagnostic across 42 conditions (0.35 h, microphones 0-3). The retained gated path produced 0.0 observed events/hour; KWS-only produced 8.57/hour. This supports hard-negative testing and future augmentation, but contains no speech, passenger, wake, or command evidence and was not added to training.

Run	Duration	Full-path wakes/h	KWS-only/h	Positive proxy	Decision
Retained reference (merged)	0.335 h	728.1/h	1817.2/h	68.8% wake / 59.4% intent	Reference; negative stress only; 105.9/h actions
Full Smart hard-negative (merged)	0.335 h	17.9/h	50.7/h	14.8% wake / 2.3% intent	Reject: positive loss; 0.0/h actions
Thin Smart candidate (legacy)	0.335 h	2461.8/h	13263.7/h	81.2% wake / 75.0% intent	Reject: no negative gain; 105.9/h actions
Thin, KWS 0.70 (legacy)	0.335 h	0.0/h	0.0/h	0.0% wake / 0.0% intent	Reject: 0% wake; 0.0/h actions
Distilled Smart candidate	-	-	-	31.2% wake / 31.2% intent	Reject: low wake; negative run not performed

The Smart Speaker dataset is clean 48 kHz mono studio audio from seven speakers saying “Hey Alexa” plus a command; it is licensed CC BY 4.0 and is not cabin or multichannel evidence. The clips were duplicated across four channels only for interface compatibility. The full candidate reduced the negative diagnostic but fell to 14.8% wake recall; the thin candidate reached 81.3% wake recall on only 16 synthetic scenes without improving the negative diagnostic. The 0.70 threshold produced no observed negative events and no positive wakes. No candidate replaces the retained reference.

Personal activity-head candidate

Presence operating point	Wake recall	Overall intent	False wakes/h	False actions/h	Decision
0.15 / command 0.10	81.2%	62.5%	208.9/h	96.4/h	Reject: unsafe
0.40 / command 0.10	50.0%	37.5%	0.0/h	0.0/h	Reject: recall loss

The new personal presence head conditions the frame gate on the SpEx+ enrollment embedding. It reaches 81.3% wake recall at the inherited threshold but produces 96.4 false actions/hour; the validation-derived 0.40 threshold removes observed proxy actions but drops wake recall to 50.0%. It remains a rejected research candidate until a real-driver tuning split exists.

Enrollment-bank candidate

Reference representation	Bank size	Wake recall	Overall intent	False actions/h	Decision
Single short utterance	1	81.2%	68.8%	0.0/h	Retain as default
Uncalibrated mean	3	6.2%	6.2%	0.0/h	Reject: recall collapse

A three-utterance SpEx embedding bank was tested as a literature-inspired runtime option. It produced zero observed proxy actions only by missing nearly every target wake (6.25% recall), so it is not promoted. The mechanism remains available for same-driver tuning with quality-weighted or learned aggregation.

Late speaker-verification fusion candidate

Candidate	Proxy wake	Proxy intent	Smart full-path/h	Decision
Min, center .24	100.0%	100.0%	728.1	No improvement
Min, center .26	87.5%	87.5%	695.3	Reject: recall loss
Product, center .24	25.0%	25.0%	not run	Reject: severe recall loss

The opt-in late-fusion path combines target-KWS evidence with a sigmoid-mapped SpEx cosine score before smoothing. Min fusion at cosine center 0.24/scale 0.04 retained 100.0% proxy wake/intent and 0 false actions but matched the retained Smart-negative rate; center 0.26 reduced that rate only to 695.3/hour while dropping proxy recall. Product fusion dropped proxy recall to 25.0%. The raw SpEx similarity is therefore not calibrated enough for promotion; the reference path remains unchanged. Detailed evidence is in reports/late_speaker_fusion_candidate_2026-10-11.md.

12A. Runtime-domain speaker-verifier calibration audit

The literature review suggested a separate text-independent speaker branch. Before collecting the real-driver pilot, the frozen SpEx enrollment and extracted-mixture embeddings were calibrated on the existing Stage-3 synthetic CAVEMOVE/GSC-Marvin target-present/target-absent cache. This is closer to the runtime path than clean speaker pairs, but it is still not real-driver or real-passenger evidence.

Measurement	Result
Calibration rows	1,208 train / 260 validation / 256 test
Held-out test AUC	0.515
Validation operating point	0.0% TPR / 0.8% FPR
Held-out test at selected point	1.6% TPR / 0.8% FPR
End-to-end proxy wake / intent	56.2% / 43.8%
End-to-end proxy false actions	0.0/h

The fitted mapping was sigmoid((cosine - 0.2661) / 2.7197). The held-out AUC of 0.515 is chance-level; at the validation point chosen for about 1% FPR, test TPR was only 1.56%. In the 16-condition proxy, the candidate retained 56.25% wake recall and 43.75% overall intent with zero observed false events. The zero-negative result is caused by suppressing target events and is not a usable safety result.

Decision: reject the calibration-only late-fusion candidate. The evidence points to an upstream representation problem, not a missing sigmoid threshold. A production-oriented upgrade needs a separately trained speaker-conditioned activity/verifier head using permitted same-driver enrollment, driver wake/command, passenger speech, passenger Marvin, and no-command intervals, with thresholds frozen on tuning sessions before held-out evaluation.

Detailed evidence: reports/speaker_verifier_calibration_2026-10-11.md

Pretrained speaker-encoder follow-up

Verifier	Test AUC	Test TPR	Test FPR
SpEx extracted embedding	0.515	1.6%	0.8%
WavLM cosine	0.839	52.3%	7.0%
Frozen WavLM + relation head	0.896	37.5%	2.3%

WavLM improves representation quality but is not an edge-ready branch: one 1-second forward measured 25.7 ms p50 / 31.6 ms p95, with 101.1M parameters and 443.8 MB peak CUDA memory. It remains a teacher/diagnostic candidate pending same-driver data, distillation, quantization, and target-device measurements.

Compact verifier runtime candidate

Measurement	Result
Held-out test AUC	0.917
Test EER	15.6%
TPR / FPR at validation point	16.4% / 2.3%
Parameters	176,579
RTX 4080 p50 / p95	5.14 / 5.24 ms
Proxy wake / intent at relaxed 0.80	6.2% / 0.0%

The compact verifier is a fast, memory-light 176,579-parameter late speaker veto, but its low-FPR operating point transferred poorly. The relaxed full-cascade diagnostic recovered 1/16 wakes and 0/16 correct commands, so it is rejected for promotion. Same-driver data and causal speaker-conditioned training are required before revisiting it.

ANIR noise-augmentation follow-up

Candidate	Wake	Intent	False actions/h
Existing personal head	81.2%	62.5%	96.4/h
ANIR-augmented personal head	81.2%	62.5%	96.4/h

ANIR contributes 42 matched four-channel, 16-kHz automotive-noise beds, but this augmentation did not change the 16-scene proxy result. The checkpoint is rejected as a production upgrade; the cache is retained for future same-driver training.

Three-way personal-activity follow-up

Operating point	Wake	Intent	False wakes/h	False actions/h	Decision
Binary personal head, 0.15	81.2%	62.5%	208.9/h	96.4/h	Reference
3-way activity, 0.25	75.0%	56.2%	208.9/h	48.2/h	Diagnostic; reject
3-way activity, 0.30	25.0%	18.8%	208.9/h	80.4/h	Recall loss
3-way activity, 0.40	0.0%	0.0%	0.0/h	0.0/h	Reject; all suppressed

The opt-in head predicts non-speech, target speech, and non-target speech and reached internal validation chunk F1 0.697 at 0.40. On the cross-corpus proxy, 0.25 halved observed false actions from 96.4/h to 48.2/h, but false wakes remained 208.9/h and target recall fell. It is retained for training on real passenger labels, not promoted.

On the 250-file Smart Speaker non-Marvin diagnostic, the same point changed merged full-path events from 728.1/hour to 674.4/hour but left the 14-file action sample at 105.9 false actions/hour. This is only a modest clean-studio change and does not meet the safety bar.

A speaker-disjoint retrain used 1,925 training rows across 912 enrollment groups and 483 validation rows across 225 groups, with zero overlap. Its internal chunk F1 was 0.755 at 0.45, but the proxy returned 75.0% wake / 56.2% intent / 96.4/h false actions/hour at 0.25 and rejected all scenes at 0.35. It is retained as the more honest training baseline, not promoted.

AISHELL-5 target-absent follow-up

Exposure	Full-path events/h	KWS-only events/h
3.889 h / 18 sessions	4.9/h	96.7/h

Across 3.889 hours of real four-channel AISHELL-5 target-absent speech, the group-split three-way candidate produced 4.9/h full-path events/hour versus 96.7/h KWS-only events/hour at KWS 0.50. The aggregate full-path sweep (KWS threshold->events/hour) was 0.30->11.8/h, 0.40->5.7/h, 0.50->4.9/h, 0.60->3.1/h; the presence sweep (presence threshold->events/hour) was 0.10->9.5/h, 0.15->8.7/h, 0.20->6.9/h, 0.25->4.9/h, 0.30->1.8/h, 0.35->0.8/h. This supports useful rejection on real cabin audio, but contains no same-driver positives and is not a production false-accept or recall claim.

The opt-in target-minus-passenger margin at -0.67 reduced the same target-absent full-path exposure to 3.6/h versus 4.9/h without changing the KWS-only rate (96.7/h). Because the margin was selected on the synthetic ReMASC stress proxy and there are no positive driver sessions, this is a negative-rejection diagnostic only.

12B. Pre-existing systems reference and ReMASC vehicle stress

The reference review converges on a small streaming keyword branch plus an independently calibrated speaker-conditioned activity or verification veto. Personal VAD 2.0 and target-speaker VAD motivate the three-way activity head; personalized-keyphrase systems motivate separating keyword evidence from speaker identity; multichannel hotword work motivates ANC or spatial features once matching four-channel recordings exist; and production KWS toolkits treat temporal state, long negative exposure, and the threshold/refractory policy as part of the system. These references support a data-and-calibration upgrade, not an immediate replacement of the intent model.

Configuration	Wake	Intent	False wakes/h	False actions/h	Decision
Retained frame gate	100.0%	90.6%	257.1/h	64.3/h	Reference diagnostic
3-way activity, p=0.25	100.0%	96.9%	257.1/h	128.6/h	Reject; passenger actions
3-way + margin -0.67	90.6%	78.1%	64.3/h	0.0/h	Diagnostic; no promotion
target-top margin -0.30	50.0%	28.1%	64.3/h	0.0/h	Reject; recall loss
framewise-top margin -0.30	68.8%	40.6%	64.3/h	0.0/h	Reject; recall loss
fast stream, 0.25 s / delay 4	96.9%	81.2%	64.3/h	0.0/h	Diagnostic; no promotion
3-way activity, p=0.35	0.0%	0.0%	0.0/h	0.0/h	Reject; all suppressed
p=0.25 / command=0.35	100.0%	0.0%	192.9/h	192.9/h	Reject; no commands

A 60-second, six-channel ReMASC replay was resampled and tiled as a real-vehicle stress bed, then mixed into 32 synthetic target scenes (448 seconds) at -5/0/5/10 dB cabin SNR. The retained path reached 100.0% wake and 90.6% intent with 64.3/h diagnostic false actions/hour. The group-split 3-way head improved intent to 96.9% at p=0.25 but raised false actions to 128.6/h; p=0.35 rejected every target scene. The asymmetric gate also failed. This is synthetic target-plus-real-cabin stress, not real-driver evidence or a production FAR.

The margin sweep was -0.70/-0.68/-0.67/-0.66. At -0.67 it reached 90.6% wake, 78.1% intent, 64.3/h diagnostic false wakes/hour, and 0.0/h diagnostic false actions/hour. The -0.68 point restored false actions, while -0.66 reduced wake recall to 62.5%. The cutoff is therefore retained as an opt-in research diagnostic, not a frozen operating point.

A frame-alignment follow-up compared target and passenger probabilities on the same frames. At a proxy-selected -0.30 cutoff, target-top reached 50.0% wake / 28.1% intent and framewise-top reached 68.8% / 40.6%; both had 0.0/h observed false actions/hour. Target-top reduced all-session AISHELL target-absent events to 3.3/h/hour, only slightly below the independent-top margin result. Both modes are rejected for promotion because of positive proxy recall loss.

A streaming-state follow-up reduced the hop from 1.0 s to 0.25 s and delayed intent capture by four calls so the command window no longer overlaps the wake word. The fast candidate reached 96.9% wake, 81.2% intent, 64.3/h diagnostic false wakes/hour, 0.0/h false actions/hour, and 0.968 s mean wake latency (1.625 s p95). This improves the one-second reference latency, but remains a proxy result that must be calibrated on real drivers and measured on the target CPU/DSP.

The fast setting was then checked on 1,200 seconds of real four-channel AISHELL-5 target-absent audio. Both eight-call and sixteen-call refractory settings produced 6.0/h full-path events/hour, versus 3.0/hour for the comparable one-second reference; the longer refractory changed KWS-only events but not the full path. The fast setting is therefore rejected as a current safety improvement and remains an opt-in latency study only.

Decision: keep the 3-way head as a research implementation, but do not promote it. The next experiment is permitted same-driver continuous data with target/passenger/non-speech labels, a compact speaker-verification branch, and a two-factor action veto calibrated on tuning sessions and frozen before held-out testing. The full reference list and exact commands are in reports/preexisting_systems_review_2026-10-11.md and reports/research_journey.md.

13 - Reproducibility

```
# From the repository root
python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_temporal2_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 0 --output logs/continuous_ablation_matrix_2026-10-09.json

python scripts/eval/eval_threshold_ablation_sweep.py `
  --config configs/eval_end_to_end_cavemove_target_kws_temporal2_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 0 --presence-thresholds 0.30 0.50 0.70 0.90 `
  --kws-thresholds 0.30 0.40 0.50 0.60 `
  --variants full no_speaker_matching no_wake_smoothing `
  --output logs/threshold_ablation_sweep_2026-10-09.json

python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_scalar_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 0 --output logs/continuous_ablation_matrix_scalar_presence_2026-10-09.json

python scripts/eval/eval_runtime_ablation_sweep.py `
  --config configs/eval_end_to_end_cavemove_target_kws_temporal2_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 0 --hops 0.5 1.0 --enrollment-seconds 0.5 1.0 1.5 `
  --output logs/runtime_ablation_sweep_2026-10-10.json

python scripts/eval/eval_stage1_dual_domain_regression.py `
  --max-per-domain 144 `
  --output logs/stage1_dual_domain_regression_full_2026-10-10.json

python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_dual_domain_stage2_operating_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 10 `
  --output logs/continuous_ablation_matrix_dual_domain_stage2_intent_handoff_2026-10-10.json

python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_dual_domain_stage2_operating_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 10 --enrollment-source same_speaker_other `
  --output logs/continuous_ablation_matrix_dual_domain_stage2_same_speaker_other_2026-10-10.json

python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_dual_domain_stage2_hardneg_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 10 `
  --output logs/continuous_ablation_matrix_dual_domain_stage2_hardneg_2026-10-10.json

python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_dual_domain_stage2_operating_candidate.yaml `
  --output logs/continuous_ablation_matrix_stage2_intent_confidence_gate_passenger_m10_2026-10-10.json `
  --scenes 2 --cabin-snr-db -5 0 10 20 --passenger-snr-db -10 `
  --enrollment-source same_speaker_other --speaker-gate-thresholds 0.22 `
  --intent-confidence-thresholds 0.50 0.53 0.55 0.60
```